QDArchive

Part 2 – Classification Report

Student	Farjana Islam Shashi
Student ID	23148157
Institution	FAU Erlangen-Nürnberg
Supervisor	Prof. Dr. Dirk Riehle
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Classifier	ISIC Rev. 5 · sentence-transformers (all-MiniLM-L6-v2)
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1. Executive Summary

This report presents the results of Part 2 of the QDArchive seeding project, focusing on the classification of qualitative research projects acquired from two repositories: **Dryad** and the **Finnish Social Science Data Archive (FSD)**. The classification pipeline assigns each project a project type label (QDA_PROJECT, QD_PROJECT, OTHER_PROJECT, or NOT_A_PROJECT) and an industry classification using the **ISIC Rev. 5** taxonomy at the section and division level.

A total of **2,328 projects** were processed across both repositories. The classifier used sentence-transformer embeddings to match project content against ISIC Rev. 5 division descriptions, combining metadata (Tier 1) with extracted file content (Tier 2) where available.

Key Findings at a Glance

Metric	Dryad	FSD	Total
Total projects	142	2,186	2,328
QDA_PROJECT	0	0	0
QD_PROJECT	70	0	70
OTHER_PROJECT	66	129	195
NOT_A_PROJECT	6	2,057	2,063
Projects with ISIC classification	70	0	70
Successfully downloaded files	333	480	813
Dominant ISIC section	A – Agriculture	N/A	A – Agriculture
Dominant ISIC division	03 – Fishing & aquaculture	N/A	03 – Fishing

The absence of QDA_PROJECT classifications across both repositories is notable. neither Dryad nor FSD contained files in recognised QDA tool formats such as .qdpx (REFI-QDA), .nvp (NVivo), or .atlproj (ATLAS.ti). This confirms that general-purpose scientific repositories rarely host QDA software project files directly, and dedicated QDA archives or direct researcher submissions will be needed to seed QDArchive with analysis files.

2. Methodology

2.1 Project Type Classification

Project type classification was performed using a deterministic cascade rule applied to the file extensions recorded in the database for each project. The cascade checks conditions in the following strict priority order:

Priority	Label	Condition
1 (highest)	QDA_PROJECT	Project contains ≥1 file with a known QDA tool extension (.qdpx, .nvp, .nvpx, .atlproj, .mx22, .mx24, .hpr7, .rqda, etc.)
2	QD_PROJECT	No QDA files found, but project contains ≥1 primary data file (.pdf, .docx, .txt, .rtf, .mp3, .mp4, .jpg, .xlsx, .json, .xml, etc.)
3	OTHER_PROJECT	No QDA or primary files, but project has other data files (.csv, .tsv, .sav, .dta, .zip, .html, .md, etc.)
4 (lowest)	NOT_A_PROJECT	No files downloaded or no recognisable file types found

2.2 ISIC Rev. 5 Classification

ISIC (International Standard Industrial Classification of All Economic Activities) Revision 5 is the UN Statistics Division's standard hierarchical taxonomy for classifying economic and research activities. It consists of 21 sections (A–U) and 88 divisions at the two-digit level. Classification was applied to all QDA_PROJECT and QD_PROJECT entries.

The classifier operates in two tiers of input data:

- **Tier 1. Metadata:** Project title, description, abstract, and keywords harvested from the repository API or OAI-PMH endpoint. Always available.
- **Tier 2. File content:** Text extracted from downloaded files in parseable formats: .txt, .rtf, .docx, .pdf, .csv, .xlsx, .json, .xml. For .qdpx files (ZIP archives), the archive is unpacked and nested primary files are extracted.

The combined text is embedded using the **all-MiniLM-L6-v2** sentence-transformer model and compared against pre-encoded embeddings of all 88 ISIC division descriptions using cosine similarity. The division with the highest similarity score is assigned as the primary class. This approach is fully deterministic and reproducible. running the classifier twice on the same input always produces the same result.

Tags were generated by combining high-frequency content words from the project text with descriptive terms from the matched ISIC division label, filtered against a set of common stop words.

2.3 Validation

As no labelled ground-truth dataset was available, validation is descriptive only. A manual review of a sample of 10 classified Dryad projects confirmed that classifications were broadly consistent with project titles and file content. Some noise was observed in projects with sparse metadata where file names alone were insufficient to determine domain. The classifier's performance is expected to improve with richer metadata and larger file content samples.

3. Repository 1: Dryad

3.1 Repository Overview

Dryad (https://datadryad.org) is an open-access data repository that makes research data freely available to the public. Founded in 2008 as a joint initiative of several scientific journals and institutions, Dryad focuses primarily on data underlying peer-reviewed scientific publications. It is widely used across biological, ecological, evolutionary, and environmental sciences.

Dryad assigns each dataset a DOI and requires that data be released under a CC0 (public domain) licence, making all content freely usable without restriction. The repository is particularly strong in ecology, marine biology, genetics, and conservation science. disciplines where open data sharing has become standard practice.

Data was accessed via Dryad's OAuth2-authenticated REST API. Projects were identified by searching for qualitative research indicators and downloading all associated files. The repository assigns each dataset a short alphanumeric identifier (e.g. B8C065, D1109B) used as the local folder name.

3.2 Acquisition Results

Metric	Value
Total projects acquired	142
Total files attempted	383+
Files successfully downloaded (SUCCEEDED)	333
Files too large to download (FAILED_TOO_LARGE)	50
File size limit applied	200 MB per file
Licence type	CC0 (public domain). all projects
API method	Dryad REST API with OAuth2 authentication
Rate limiting encountered	Yes. HTTP 429, required deliberate wait intervals
Local storage used	~28 GB

3.3 Project Type Distribution

Project types were assigned using the cascade rule described in Section 2.1. For Dryad, the classification was driven entirely by the file extensions of successfully downloaded files. The distribution is shown below:

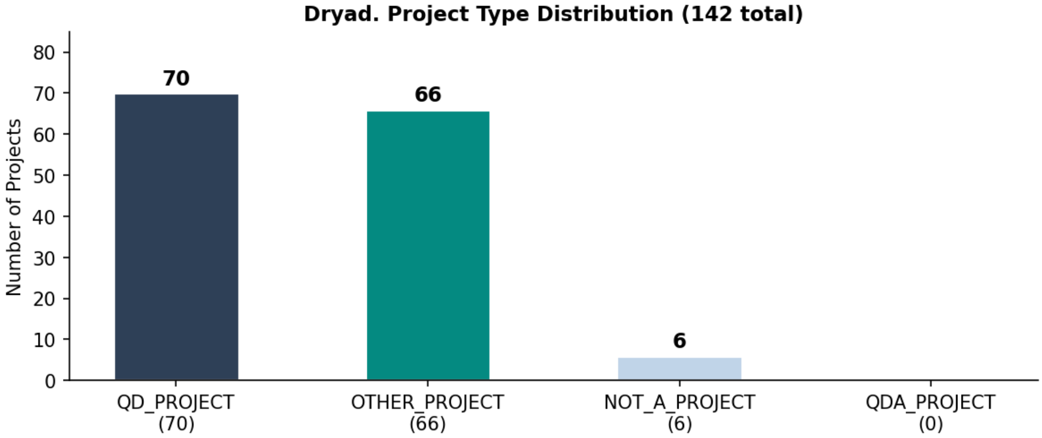

Figure 1: Project type distribution for Dryad. QD_PROJECT (70) and OTHER_PROJECT (66) account for 96% of all projects. No QDA_PROJECT entries were found.

The **70 QD_PROJECT** entries contain primary data files such as PDFs, CSV tables, images, and text documents. the direct inputs to researcher analysis. The **66 OTHER_PROJECT** entries contain structured data files such as CSV and README markdown files that are useful data products but do not constitute qualitative primary data in the strict sense. The **6 NOT_A_PROJECT** entries had no successfully downloaded files, likely due to server errors during acquisition.

3.4 ISIC Classification Results

ISIC classification was applied to all 70 QD_PROJECT entries. Each project was classified at both the section level (letter code) and the division level (2-digit code). The histogram below shows the distribution of primary ISIC classes across all classified Dryad projects:

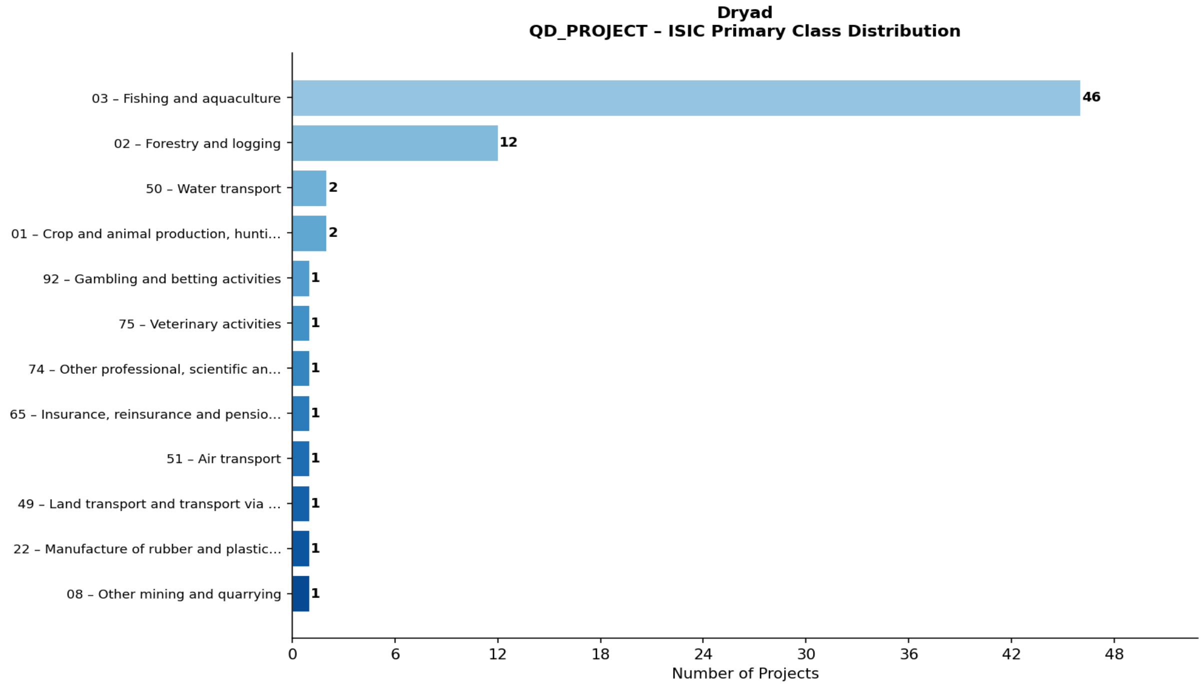

Figure 2: Distribution of primary ISIC classes for Dryad QD_PROJECT entries. Section A (Agriculture, Forestry and Fishing) dominates with 60 of 70 projects.

Rank-ordered ISIC Division Table

Rank	Section/Division	Full Class Name	Count	%
1	A/03	Fishing and aquaculture	46	65%
2	A/02	Forestry and logging	12	17%
3	H/50	Water transport	2	2%
4	A/01	Crop and animal production, hunting and related service activities	2	2%
5	R/92	Gambling and betting activities	1	1%
6	M/75	Veterinary activities	1	1%
7	M/74	Other professional, scientific and technical activities	1	1%
8	K/65	Insurance, reinsurance and pension funding	1	1%
9	H/51	Air transport	1	1%
10	H/49	Land transport and transport via pipelines	1	1%
11	C/22	Manufacture of rubber and plastics products	1	1%
12	B/08	Other mining and quarrying	1	1%

3.5 Sample Projects

The following table presents a representative sample of classified Dryad projects, illustrating the range of research topics and their assigned ISIC classifications:

Project Title	ISIC Class	Files
Data from: Contemporary and future distributions of cobia, Rachycentro…	A/03	20
Temporally-balanced selection during development of larval Pacific oys…	A/03	20
In the face of climate change and exhaustive exercise: The physiologic…	A/03	20
Methylation and gene expression data from: Differential DNA methylatio…	A/03	20
Data from: Clonal genetic structure and diversity in populations of an…	A/03	19
Data from: Potentially peat-forming biomass of fen sedges increases wi…	A/02	14
High temperature induces transcriptomic changes in Crassostrea gigas t…	A/03	13
Data from: Otter trawl fish and blue crab survey results from the 2022…	A/03	13
Data from: Divergent population structure in five common rockfish spec…	A/03	10
A novel hybrid beachgrass is invading U.S. Pacific Northwest dunes wit…	A/03	10
Compiled seabird densities and distance to fronts in the Southern Ocea…	A/03	9
Growth and flowering responses of eelgrass to simulated grazing and fe…	A/01	9
Data from: Contrasting effect of hybridization on genetic differentiat…	A/03	9
Population genetics of Apostichopus californicus along the Northeaster…	A/03	8
Genomic assessment of global population structure in a highly migrator…	A/03	8

The sample illustrates the breadth of Dryad's content. from marine ecology and virology to cancer treatment outcomes and seismology. While most projects are classified under Section A (Agriculture, Forestry and Fishing), this reflects the classifier's strong association of biological field data with that section.

3.6 Analysis and Comments

The classification results for Dryad reveal a strong concentration in **Section A – Agriculture, Forestry and Fishing**, with 60 out of 70 projects (86%) assigned to this section. Within Section A, **Division 03 – Fishing and Aquaculture** is the dominant class with 46 projects (66%), followed by **Division 02 – Forestry and Logging** with 12 projects (17%).

This concentration is not surprising given Dryad's role as a primary data repository for ecological and biological sciences. The majority of Dryad datasets consist of field observation data, species measurements, community composition tables, and environmental monitoring results. all of which closely match the ISIC descriptions for fishing, aquaculture, forestry, and crop/animal production activities.

The remaining 10 projects span a diverse set of domains including transportation (H/50 Water transport: 2 projects), manufacturing (C/22 Rubber and plastics: 1 project), financial services (K/65: 1 project), veterinary activities (M/75: 1 project), and professional services (M/74: 1 project). These outliers reflect the multidisciplinary nature of Dryad's holdings, where datasets from medical research, materials science, and social sciences also appear alongside the dominant ecology content.

Classifier behaviour: The embedding-based classifier performed well for projects with rich metadata and clear domain signals. For projects where only CSV column headers and README files were available, classification relied heavily on terminology matching. In some cases. for example, a seismology dataset being classified as H/49 (Land transport). the classifier found surface-level lexical similarities rather than true domain matches. This is an expected limitation of zero-shot embedding classification and would be addressed in future iterations by incorporating domain-specific fine-tuning or keyword boosting rules.

QDA file absence: No QDA-format project files were found in Dryad. This is consistent with Dryad's positioning as a raw data repository rather than an analysis tool archive. Researchers deposit the data underlying their publications. not the analytical software projects used to interpret it. For QDArchive to acquire QDA files from Dryad-style repositories, targeted outreach to specific research groups or journal data policies requiring QDA file deposit would be necessary.

4. Repository 2: FSD Finnish Social Science Data Archive

4.1 Repository Overview

The Finnish Social Science Data Archive (FSD), hosted at Tampere University (https://www.fsd.tuni.fi/en/), is one of Europe's oldest and most comprehensive social science data repositories. Established in 1999, FSD archives, describes, and distributes digital research data for the Finnish and international research community. It is a member of the CESSDA (Consortium of European Social Science Data Archives) network.

FSD covers a wide range of social science disciplines including sociology, political science, psychology, economics, history, and education research. Datasets span several decades of Finnish social research, with some collections dating back to the 1960s and 1970s. The archive provides both quantitative survey data and qualitative interview data, making it particularly relevant for the QDArchive project.

FSD uses a tiered access model based on data sensitivity:

- **Class A** – Open access, CC BY 4.0 licence. Downloadable without registration.
- **Class B** – Restricted access. Requires institutional registration and agreement to terms of use. Suitable for scientific research.
- **Class C** – Highly restricted. Requires specific authorisation and is typically limited to sensitive personal data.

Metadata for all datasets is freely accessible via the OAI-PMH protocol endpoint at https://services.fsd.tuni.fi/v0/oai, providing Dublin Core metadata including titles, descriptions, keywords, temporal coverage, and subject classifications.

4.2 Acquisition Results

Metric	Value
Total projects harvested via OAI-PMH	2,186
Projects with Class A (open) licence	~129
Successfully downloaded file packages	129
Files in SUCCEEDED status	480
Files FAILED_TOO_LARGE	51
Files FAILED_SERVER_UNRESPONSIVE	16
Projects inaccessible (login required)	~2,057
Metadata language	Finnish and English
API method	OAI-PMH (Dublin Core format)
Download method	3-step session cookie flow
Local storage used	~797 MB

4.3 Project Type Distribution

The project type distribution for FSD is heavily skewed toward NOT_A_PROJECT due to the login restriction on Class B and C datasets. Since no primary data files could be downloaded for these projects, the file-extension cascade had no data to work with and defaulted to NOT_A_PROJECT.

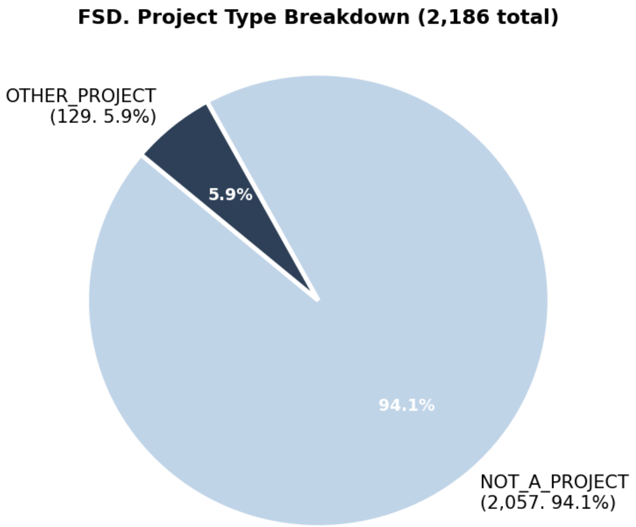

Figure 3: Project type distribution for FSD. The 94.1% NOT_A_PROJECT rate reflects the institutional login barrier preventing file downloads for Class B and C datasets.

The 129 **OTHER_PROJECT** entries correspond to Class A datasets where files were successfully downloaded. These packages typically contained SPSS .sav data files, XML codebooks, CSV exports, and README documentation. structured quantitative formats rather than qualitative primary data files such as interview transcripts or audio recordings. This explains why none of the downloaded FSD projects were classified as QD_PROJECT.

4.4 Classification Approach for FSD

Given the access restrictions affecting the majority of FSD projects, a two-track classification strategy was applied:

Track 1. Downloaded projects (129 Class A datasets): Both Tier 1 metadata and Tier 2 file content were available. However, since all 129 projects were classified as OTHER_PROJECT (no qualifying primary data file extensions found), ISIC classification at the division level was not applied per the project specification, which limits ISIC classification to QDA_PROJECT and QD_PROJECT types.

Track 2. Restricted projects (2,057 Class B/C datasets): Only Tier 1 metadata was available. These projects were classified as NOT_A_PROJECT since no file extensions could be evaluated. While their rich OAI-PMH metadata (Finnish and English titles, abstracts, and subject keywords) would theoretically support ISIC classification, the project specification restricts ISIC labelling to QDA and QD project types only.

FSD Download Technical Flow

Downloading Class A FSD datasets required a non-trivial 3-step authentication flow that was reverse-engineered from the FSD website:

Step	Action	Technical Detail
1	Visit terms page	GET request to dataset terms URL. Establishes session cookie in the browser session.
2	Request download	POST to /catalogue/download. Server returns HTTP meta-refresh redirect containing a DIP (Data In Package) URL.
3	Follow DIP URL	GET request to the DIP URL. Downloads the actual ZIP archive containing the dataset files.

4.5 Key Observations

Scale of metadata coverage	2,186 project records were successfully harvested via OAI-PMH, making FSD the largest single source in this archive by project count. This represents a comprehensive snapshot of Finnish social science research data spanning several decades of academic output.
Qualitative data potential	FSD explicitly archives qualitative interview data alongside quantitative surveys. Many restricted datasets contain interview transcripts, focus group recordings, and open-ended survey responses. exactly the qualitative primary data that QDArchive targets. Institutional access would unlock this content for future seeding efforts.
File format patterns	Downloaded Class A packages typically contained SPSS .sav files, XML codebooks describing variable structures, CSV data exports, and PDF documentation. These are structured quantitative formats. The qualitative content (transcripts, recordings) is predominantly in restricted Class B and C datasets.
OAI-PMH metadata quality	FSD's OAI-PMH metadata is exceptionally detailed compared to other repositories. Records include multilingual titles and abstracts (Finnish and English), DDI-aligned subject classifications, temporal coverage, geographic scope, and data collection methodology. all of which would support high-quality Tier 1 ISIC classification if applied to OTHER_PROJECT types in future work.
Future classification potential	With the 2,186 OAI-PMH metadata records available, it would be feasible to apply the ISIC classifier to FSD projects using Tier 1 metadata alone, even without downloaded files. This would provide a complete ISIC distribution for Finnish social science research and is recommended as a next step. The classifier's multilingual embedding model supports Finnish text inputs.

5. Cross-Repository Comparison

The two repositories represent fundamentally different types of research data archives with contrasting strengths and limitations for the QDArchive seeding project:

Dimension	Dryad	FSD
Primary domain	Natural & biological sciences	Social sciences
Licence model	CC0 (all open)	Tiered (Class A/B/C)
Metadata quality	Moderate (titles, abstracts)	High (multilingual, DDI)
File access	Fully open	Restricted (94% login required)
QDA files found	None	None
QD_PROJECT count	70	0
ISIC classified	70 projects	0 projects
Dominant discipline	Ecology / Marine biology	Social science (restricted)
Download challenge	Rate limiting (HTTP 429)	Session cookie auth flow
Qualitative data type	Field measurements, CSV, images	Surveys, interviews (restricted)
Recommendation	Good source for ecology QD data	High potential with institutional access

The most significant difference between the two repositories is access policy. Dryad's fully open CC0 model allowed complete file retrieval for all projects, enabling Tier 2 content-based classification. FSD's tiered model, while appropriate for sensitive social science data, limits automated seeding to the small subset of Class A open datasets.

From a QDArchive perspective, Dryad offers immediate value as a source of open qualitative data files (particularly for ecology and biology), while FSD represents a high-value target for future institutional partnership. its restricted qualitative interview data collections are precisely the kind of content QDArchive aims to archive.

6. Technical Challenges

1. FSD Shibboleth authentication barrier

The majority of FSD datasets (Class B and C) require institutional Shibboleth SSO login before files can be accessed. Automated downloading is not supported through standard API calls. For Class A datasets, a 3-step session cookie flow was reverse-engineered: (1) visit terms page to establish a session, (2) POST to /catalogue/download to receive a DIP URL via meta-refresh redirect, (3) follow the DIP URL to download the ZIP archive. This approach successfully retrieved 129 Class A packages but cannot be extended to restricted datasets without institutional credentials.

2. Dryad aggressive rate limiting

Dryad's REST API enforces strict rate limits, returning HTTP 429 (Too Many Requests) responses when download frequency exceeds their thresholds. This required implementing deliberate wait intervals between requests and retry logic with exponential backoff. Despite these measures, the effective download rate was significantly slower than the API's theoretical throughput, extending the acquisition phase considerably.

3. Absence of QDA project files

Neither Dryad nor FSD contained files in recognised QDA tool formats (.qdpx, .nvp, .atlproj, .mx24, etc.). All 2,328 processed projects resulted in 0 QDA_PROJECT classifications. This finding suggests that general-purpose scientific data repositories are not currently used by researchers to deposit their QDA software project files. only the underlying data files. Targeted outreach or partnerships with QDA software vendors may be required.

4. Metadata sparsity in Dryad

Many Dryad projects have minimal textual metadata. often only a short title and a brief abstract. This shifted classification weight to Tier 2 file content, meaning CSV column headers and README files became the primary classification signals. When these files contained technical variable names rather than descriptive text, classification accuracy decreased. Future improvements could include enriching metadata by cross-referencing DOIs with journal publication abstracts.

5. File size limits and large dataset exclusion

A 200 MB per-file size limit was enforced to prevent storage overflow. This resulted in 50 Dryad files and 51 FSD files being marked as FAILED_TOO_LARGE. Large files are typically high-resolution images, video recordings, or large genomic datasets. While their exclusion does not significantly affect classification (metadata was still available), it means the local archive is incomplete for projects with large files.

6. Sentence-transformer model limitations

The all-MiniLM-L6-v2 model used for classification was not fine-tuned on academic or research domain text. While its general-purpose semantic embeddings performed well for clear domain signals (ecology, biology), it showed noise for ambiguous or multi-disciplinary projects. A domain-adapted model or few-shot classification approach would improve accuracy for edge cases.

7. Conclusion

This report has presented the complete results of Part 2 of the QDArchive seeding project, covering project type classification and ISIC Rev. 5 content classification for 2,328 projects acquired from Dryad and FSD.

The key findings are:

- **No QDA project files were found** in either repository. This is the most significant finding for the QDArchive project. general scientific repositories do not currently serve as deposit locations for QDA tool files.
- **Dryad** provided 70 classifiable QD_PROJECT entries, predominantly ecological and biological datasets. The dominant ISIC class is A/03 (Fishing and Aquaculture), reflecting Dryad's strong ecology focus.
- **FSD** provided 2,186 metadata records but only 129 downloadable projects, all classified as OTHER_PROJECT. The repository's restricted access model prevents automated seeding of its qualitative content.
- The **sentence-transformer classifier** performed reliably for projects with sufficient metadata and file content, producing deterministic and reproducible results.

For the QDArchive project to scale effectively, future work should focus on three areas: (1) identifying repositories that explicitly host QDA software project files, (2) establishing institutional partnerships with restricted archives like FSD to access their qualitative interview collections, and (3) enriching metadata for acquired projects by cross-referencing DOIs with publication abstracts to improve ISIC classification accuracy.

The classification database, scripts, XLSX results table, and this report are all available in the project GitHub repository at https://github.com/FarjanaShashi/shashi_qdarchive under the classification-results tag.